K J Somaiya School of Engineering, Somaiya Vidyavihar University

LY Agentic AI Laboratory — Lab Evaluation

SOMAIYA AGENTREEL CHALLENGE

Autonomous AI-Agent Production of Promotional & Awareness Media for Campus, Admissions, Safety and Responsible AI

Item | Details
Course / Semester | Agentic AI Laboratory — Final Year (LY), Semester ___
Evaluation type | Lab evaluation / mini-project (Build + Live Run + Viva)
Team size | 2 students per team
Output | Promotional / awareness video (30–60 s) OR interactive digital experience
Inspiration | AGEWEC 2026 — Agentic Work Execution Challenge (agewec.com)
Faculty in-charge | Dr Pankaj Mishra

Shared FolderShared Sheet (for group details)

1. Overview

This lab evaluation is modelled on AGEWEC 2026, an international competition in which AI agents autonomously research, plan, generate, edit and publish promotional media for a city. Here the "city" is our own campus. Each team builds a multi-agent pipeline that turns a sealed brief (topic + audience + format) into a finished 30–60 second video or an interactive digital experience, with minimal human intervention and full execution logs.

Evaluation covers not only the final output but also planning, prompts, the generation process, tool orchestration, cost discipline, autonomy and reproducibility.

1.1 Learning outcomes

After completing this evaluation, students will be able to:

Design a multi-agent workflow with clearly defined roles, hand-offs and a critic/feedback loop.

Ground LLM generation in verified source documents using retrieval (RAG) to avoid hallucinated claims.

Orchestrate text, image, video, speech and audio generation tools under time and budget constraints.

Instrument an agent system with structured logs, cost tracking and one-command reproducibility.

Apply responsible-AI practices: factual grounding, consent, disclosure, safe messaging and licensing.

1.2 Why this matters to the institution

Admissions & PR office receives a pool of ready-to-review promotional and awareness content.

Safety and awareness outputs (anti-ragging, ICC/POSH, cyber safety, AI policy) are directly usable in student induction and support UGC / NAAC compliance activities.

Logs and cost reports make every output auditable, answering "how was this AI content made?"

Students graduate with a real-client portfolio piece and a reproducible agentic system on GitHub.

2. Mission and Format

MISSION Build an AI-agent pipeline that autonomously researches, scripts, generates, edits and quality-checks a promotional or awareness video (30–60 s) or a digital experience for Somaiya, based on a sealed brief drawn on the day of the Live Run. AI agents must do the work. Students design, orchestrate and supervise.

2.1 Three phases

Phase | Duration | What happens
1. Build | 2 weeks (off-lab + lab hours) | Build a reusable, brief-agnostic pipeline. Ingest the official fact sheet & media kit. Dry-run on the practice brief (Section 11.4). Push to GitHub.
2. Live Run | 3 hours (one lab slot) | Each team draws a sealed brief card. The pipeline must produce the final output

within the slot, with logging on. Human interventions are counted.
3. Viva & Demo | 5–7 min per team | Show output, walk through logs and agent graph, re-run one stage live, answer questions (Section 9.5).

2.2 Suggested timeline

When | Milestone
Week 0 | Handout released; teams registered; fact sheet & media kit shared by department / admissions office.
Week 1 | Architecture & agent-graph review (5-min check-in with faculty). RAG over fact sheet working. Script + storyboard agents working.
Week 2 | Asset generation, editor and critic loop integrated. Full dry-run on practice brief. Repo frozen at tag v1.0-build 24 h before Live Run.
Live Run day | Brief draw → run → submission at T+3h (hard stop).
Viva day | Demo, logs review, live re-run of one stage, marks finalised.
After evaluation | Top entries reviewed by PR/admissions office for official use.

3. How To Do It — Step-by-Step

3.1 Reference pipeline

Teams may add agents or change the framework, but the following roles are the minimum. Every agent must log its inputs, outputs, model, tokens and cost.

Agent | Responsibility | Typical output
1. Planner | Parse brief.yaml (topic, audience, format, tone, duration). Decompose into tasks, allocate time & budget per stage. | plan.json with stages, budget split, success criteria
2. Researcher (RAG) | Retrieve facts only from the official fact sheet / brochure / policy PDFs. Attach source IDs to every claim. | facts.json (claim, source, page)
3. Scriptwriter | Write a hook (first 3 s), body and call-to-action within the word budget (~130–150 words for 60 s VO). | script.md with timestamps
4. Storyboarder | Break script into 6–12 shots: visual description, on-screen text, duration, asset source (official / generated). | storyboard.json
5. Asset Generator | Generate or select images/clips, TTS voice-over, background music, subtitles. | assets/ folder + manifest
6. Editor | Assemble timeline with ffmpeg / MoviePy: transitions, captions, logo, end card, loudness normalisation. | final.mp4
7. Critic / QA | Check duration, resolution, every factual claim vs facts.json, helplines/contacts, brand rules, safe-messaging rules. Return pass/fail with reasons; on fail, route back to the responsible agent (max 3 loops). | qa_report.json

Flow: Brief → Planner → Researcher → Scriptwriter → Storyboarder → Asset Generator → Editor → Critic → (fail: loop back | pass: export + logs).

3.2 Step-by-step procedure

Set up the repository with the mandatory structure (Section 6) and a config file for API keys (never commit keys).

Ingest the fact sheet, brochure and policy PDFs into a vector store; test retrieval with 10 sample questions.

Implement the Planner to read brief.yaml and emit plan.json with a per-stage time and cost budget.

Implement Researcher → Scriptwriter → Storyboarder with structured (JSON) outputs and schema validation.

Implement asset generation. Prefer official media-kit footage; generate only what is missing. Record licence/source for every asset.

Implement the Editor using ffmpeg or MoviePy. Burn in subtitles, add logo, end card and AI-disclosure line.

Implement the Critic with explicit, checkable rules (duration, resolution, fact match, contact verification). Wire the feedback loop.

Add the logging wrapper around every model/tool call → logs/run_log.jsonl and cost_report.csv.

Make the full run reproducible with one command: python run.py --brief briefs/<id>.yaml --seed 42

Dry-run the practice brief end-to-end at least twice; measure time and cost; tag the repo v1.0-build.

On Live Run day: draw the brief, create the brief.yaml, run, log interventions in interventions.md, submit before T+3h.

At viva: demo the output, walk through the logs, re-run one stage on request.

3.3 Suggested tool stack (not mandatory)

Layer | Options
Agent framework | LangGraph, CrewAI, AutoGen / AG2, OpenAI Agents SDK, Claude Agent SDK, or custom Python orchestrator
LLMs | Any hosted API (Claude, GPT, Gemini) or local open models via Ollama / vLLM (Llama, Qwen, Mistral)
RAG | Chroma / FAISS + sentence-transformers or API embeddings; PyMuPDF for PDFs
Images | Official media kit first; then open image models (SDXL / FLUX via diffusers) or image APIs
Video | Ken Burns / parallax on stills via MoviePy; image-to-video or text-to-video APIs if within budget
Voice-over | Edge-TTS, Piper, Coqui XTTS or TTS APIs (English / Hindi / Marathi allowed)
Music | Royalty-free libraries (e.g. YouTube Audio Library, Pixabay Music) or AI music with a usable licence
Subtitles | Whisper / faster-whisper on the generated VO → SRT → burned in
Editing | ffmpeg, MoviePy; loudnorm filter for audio levels
Digital experience | Single-page HTML/JS, Streamlit or Gradio; deploy on GitHub Pages / Hugging Face Spaces

3.4 Autonomy levels (declare one in README)

Level | Description | Human role
L1 | Assisted generation | Human writes script/storyboard; AI only generates assets
L2 | Supervised pipeline | AI plans and scripts; human approves every stage
L3 | Checkpointed autonomy | End-to-end AI; human reviews at ≤2 checkpoints
L4 | Full autonomy | One command; AI self-critiques and repairs; human only starts the run

4. Constraints

Constraint | Rule
Live Run time | 3 hours hard stop from brief draw to submission. Late = penalty (Section 9.3).
API budget | Maximum ₹500 (≈ USD 6) per Live Run across all paid APIs. Local/open models count as ₹0 but must be logged. Build-phase spend is not evaluated but should be reported.
Human interventions | Maximum 5 during Live Run. Each must be logged in interventions.md (time, stage, what was changed, why). Editing a generated file by hand = 1 intervention.
Facts & claims | Only from the official fact sheet, brochure, policy documents and notices provided. No web-scraped statistics unless the source URL is an official Somaiya / government page and is logged.
Assets | Official media kit, self-shot footage with written consent, or AI-generated assets. No copyrighted film/TV clips, celebrity images or unlicensed music.
Pre-made content | Scripts, storyboards or final media prepared before the brief draw are not allowed. Code and prompt templates are allowed.
Logging | Logging must be on for the entire Live Run. Missing or edited logs = zero for Workflow & Reproducibility.

5. Output Specifications

5.1 Video track

Parameter | Specification
Duration | Minimum 30 s, maximum 60 s (including end card). Over 60 s = penalty.
Aspect ratio & resolution | 9:16 vertical, 1080 × 1920 (Reels / Shorts) — default or 16:9 horizontal, 1920 × 1080 (YouTube / display screens) — if the brief says so
Frame rate | 25 or 30 fps (constant)
Container & codecs | MP4, H.264 (High profile), yuv420p; audio AAC, 48 kHz, stereo, 128–192 kbps
File size | ≤ 100 MB
Audio loudness | Integrated loudness about −14 LUFS; no clipping; voice clearly above music

Subtitles | Mandatory burned-in captions (readable on mobile, min. ~48 px at 1080 width); also submit .srt
Language | English by default; Hindi / Marathi or bilingual allowed if the brief permits
Hook | Topic must be clear within the first 3 seconds
End card (3–5 s) | Official logo, call-to-action (website / helpline / registration link from fact sheet), and the line "Created with AI agents — LY Agentic AI Lab"
Safe zones | Keep text away from the top 250 px and bottom 350 px in 9:16 (platform UI overlays)
File naming | TeamID_BriefID_final.mp4 (e.g. T07_A03_final.mp4)

5.2 Digital experience track

Parameter | Specification
Form | Single-page web app, chatbot, quiz, or interactive tour — completable by a user in ≤ 60 s for the core flow
Deployment | Public URL (GitHub Pages / Hugging Face Spaces / Streamlit Cloud) + source in repo
Responsiveness | Works on a mobile browser (360 px width) and desktop
Grounding | Any chatbot must answer only from the provided documents and cite the source section; must refuse out-of-scope questions
Demo video | A ≤ 60 s screen recording in the video specs above, showing the core flow
Disclosure | Visible "AI-generated / AI-assisted" label and link to the official source documents

6. Deliverables and Repository Structure

Submit a GitHub repository link (public or shared with the faculty account) tagged v1.0-final, plus the final output uploaded to the class drive.

```
agentreel-<teamID>/
 README.md            # overview, autonomy level, how to run, results
 run.py               # one-command entry point
 requirements.txt     # pinned versions
 .env.example         # key names only, never real keys
 briefs/              # brief.yaml files (practice + live)
 agents/              # planner.py, researcher.py, scriptwriter.py, ...
 prompts/             # every system/user prompt template, versioned
 data/                # fact sheet, brochure, policies (as provided)
 assets/              # generated + official assets with manifest.csv (source, licence)
 outputs/             # final.mp4 / .srt / qa_report.json / screenshots
 logs/
   run_log.jsonl      # one line per model/tool call
   cost_report.csv    # stage, provider, model, tokens/units, cost
   interventions.md   # every human intervention
 docs/
 agent_graph.png    # architecture diagram
 process_report.pdf # 1–2 pages
```

6.1 brief.yaml example

```
id: A03
category: Campus Safety & Wellbeing
topic: Women's safety and ICC/POSH — who to contact
audience: first-year students and parents
```

format: video_9x16 # or video_16x9 / digital_experience

duration_s: 45

tone: reassuring, clear, non-alarming

language: English with Hindi subtitles

must_include: [ICC contact from fact sheet, confidentiality message]

must_avoid: [depiction of victims, fear-based imagery]

cta: "Reach the ICC — details on the college website"

6.2 run_log.jsonl — one record per call

{"ts":"2026-10-12T10:14:03","run_id":"T07-A03-01","agent":"scriptwriter",

"tool":"llm","model":"<model-name>","prompt_id":"script_v3",

"input_tokens":1840,"output_tokens":412,"cost_inr":0.92,"latency_s":6.1,

"status":"ok","output_path":"outputs/script.md","output_sha256":"9f2c...",

"human_intervention":false}

6.3 Process report (1–2 pages)

Brief interpretation and creative strategy (audience insight, hook, CTA).

Agent graph and why each agent exists; framework and models chosen.

Time and cost per stage (table from cost_report.csv).

Critic loop: what failed, what the agents fixed automatically.

Human interventions and how the pipeline could remove them next time.

Limitations, ethical considerations and what you would improve.

7. Integrity, Safety and Ethics Rules

7.1 Rules for all briefs

No fabricated facts. No invented rankings, placement numbers, package figures, awards or accreditation claims. Every factual claim must trace to facts.json.

No real likeness. Do not generate the face or cloned voice of any real faculty member, student, alumnus or public figure.

Consent. Self-shot footage of identifiable people requires written consent (form in the repo).

Licensing. Only royalty-free / properly licensed music, fonts and footage; record the licence in assets/manifest.csv.

Brand. Use the official logo as provided without distortion or recolouring; no parody of the institution.

Disclosure. Every output carries the AI-disclosure line on its end card or label.

API keys & data. Never commit keys or personal data. No student personal data in prompts.

Academic honesty. Code borrowed from tutorials/repos must be credited in the README.

7.2 Additional rules for Category A (Safety & Wellbeing) and Category C (Responsible AI)

Sensitive-topic rules • Safe messaging: mental-health content must never depict or describe self-harm or suicide methods; tone must be supportive and hopeful, never fear-based; must end with a help/helpline card. • Verified contacts only: every helpline number, committee name, email and procedure must come from the fact sheet. The Critic agent must explicitly check each one. A wrong helpline or contact = zero for Accuracy. • No victim portrayal: no generated faces of "victims" or real students; use abstract, illustrated or silhouette visuals. • Mandatory human review checkpoint before export (does NOT count toward the 5-intervention limit). • No blame or shame: do not portray any gender, community, region or group as perpetrators. • Legal accuracy: describe laws and policies (UGC anti-ragging regulations, POSH Act,

DPDP Act, IT rules) only as stated in the provided documents.

8. Sealed-Brief Topics

Topics are printed as brief cards (Section 11.3) and drawn by lot on the Live Run day. Faculty may restrict the draw by category to balance coverage. Each card specifies audience, format and duration.

A. Campus Safety & Wellbeing

ID | Topic | Primary audience
A01 | Zero-tolerance anti-ragging: how to report, the national anti-ragging helpline, and what happens after a complaint | Freshers & parents
A02 | "You're not alone": mental-health support on campus — counsellor, mentor system, Tele-MANAS 14416 | All students
A03 | Women's safety and ICC/POSH: who to contact, confidentiality assured | Students & parents
A04 | Drug and substance awareness: saying no, and where to get help | All students
A05 | Cyber safety: sextortion, "digital arrest" scams, UPI and fake-job frauds; cybercrime helpline 1930 | All students
A06 | Deepfake and morphed-image awareness: what to do if your photo is misused | All students
A07 | Lab, fire and monsoon safety drill explainer | Freshers
A08 | Equity and inclusion: anti-discrimination cell and accessibility support | All students

B. Admissions — Trust & Transparency

ID | Topic | Primary audience
B01 | Beware of fake admission agents and "management quota" scams — only official channels | Aspirants & parents
B02 | CAP rounds explained in 60 s: steps, documents, dates (official notice only) | Aspirants
B03 | Scholarships and fee-waiver schemes students often miss | Aspirants & parents
B04 | NEP explainer: majors/minors, honours, APAAR/ABC ID, multiple entry–exit | Aspirants & freshers
B05 | Fee refund and cancellation policy explained clearly | Parents
B06 | "Why Somaiya?" — separate cuts for aspirants and for parents | Aspirants / parents

C. Responsible AI Usage

ID | Topic | Primary audience
C01 | College AI policy explainer: what is allowed vs not allowed in assignments and exams | All students
C02 | "AI as tutor, not ghostwriter": academic-integrity campaign | All students
C03 | Your data, your rights: DPDP Act basics for students (consent, sharing, apps) | All students
C04 | Spot the fake: recognising AI-generated misinformation (quiz-style) | All students
C05 | AI-ready graduate: how the curriculum and labs build industry AI skills | Aspirants & recruiters

D. Festivals & Events

ID | Topic | Primary audience
D01 | Tech-fest hype teaser with a "safe fest" message (passes, no substance use, help desk) | Students across colleges
D02 | Hackathon / convocation / alumni-meet invite | Students / alumni

E. Institutional Image

ID | Topic | Primary audience
E01 | Research, incubation and startups showcase | Aspirants, industry
E02 | Placements and industry connect — a recruiter-facing pitch | Recruiters
E03 | Green campus and sustainability story | All audiences

F. Digital Experience Track (any topic above may be taken as a digital experience)

ID | Suggested experience | Linked topics
F01 | "How to report" chatbot for anti-ragging / ICC, grounded only on the official policy PDFs | A01, A03
F02 | Scam-detector quiz: is this message a fraud? (UPI, job offer, digital arrest, fake agent) | A05, B01
F03 | "Can I use AI for this?" checker based on the college AI policy | C01, C02
F04 | Admissions FAQ agent grounded on the official brochure and CAP notice | B02–B05
F05 | Interactive virtual campus tour with AI narration (clickable map) | B06, E03
F06 | "Which branch fits you?" quiz with a personalised AI summary | B06, C05

Topic balance tip: Category A and C outputs are the most useful for administration (induction, UGC/NAAC compliance). Consider ensuring at least 40% of teams draw from A or C.

9. Evaluation Rubric

Total 100 marks: Output 40 + Process 60. Scale to the lab-evaluation weightage (e.g. ÷4 for 25 marks, ÷2 for 50 marks). For Category A and C briefs, criterion O1 is assessed as "Clarity & Actionability".

9.1 Summary

Code | Criterion | Marks | Evidence
O1 | Audience appeal (A/C: Clarity & actionability) | 15 | Final output
O2 | Emotional impact & memorability | 10 | Final output
O3 | Narrative coherence & production quality | 10 | Final output
O4 | Factual, brand & safety accuracy | 5 | Output + facts.json + QA
P1 | AI autonomy | 20 | Logs + interventions.md
P2 | Workflow design & tool orchestration | 15 | Code + agent graph
P3 | Reproducibility, logging & cost discipline | 15 | Re-run + logs + cost
P4 | Technical creativity | 10 | Code + viva
Total | 100

9.2 Detailed level descriptors

Criterion | Excellent (90–100%) | Good (70–89%) | Average (40–69%) | Poor (0–39%)
O1 Audience appeal / Clarity (15) | Strong hook in 3 s; target audience obviously addressed; viewer wants to apply/attend, or (A/C) knows exactly what to do and whom to contact. | Clear message and CTA; hook or audience fit slightly generic. | Message understandable but generic; weak or late CTA. | Unclear purpose; no CTA; wrong audience.
O2 Emotional impact (10) | Memorable, shareable; tone perfectly matched (inspiring / reassuring / urgent). | Engaging with a consistent tone. | Flat; tone inconsistent in places. | Disengaging or inappropriate tone.
O3 Narrative & quality (10) | Clear arc; visuals, VO, music and captions tightly synced; within specs. | Coherent; minor sync or pacing issues. | Disjointed shots; noticeable quality issues. | Incoherent; fails technical specs.
O4 Accuracy (5) | Every claim/contact verified; brand and safety rules fully met. | One minor non-critical slip. | Several slips; no critical error. | Fabricated fact, wrong helpline, or safety rule broken.
P1 Autonomy (20) | L4 or L3 with 0–1 interventions; agents self-repair via critic loop. | L3 with 2–3 interventions. | L2; 4–5 interventions. | L1, or >5 interventions, or mostly manual.
P2 Workflow design (15) | Clean agent roles, structured I/O with schemas, RAG grounding, working critic loop, sensible model routing for cost. | All roles present; loop partial; some unstructured hand-offs. | Linear script-like chain; weak grounding. | Single prompt / no real orchestration.
P3 Reproducibility (15) | One-command run works at viva; complete logs, costs, prompts; within budget and time. | Runs with minor fixes; logs mostly complete. | Partial re-run; logs incomplete. | Cannot re-run; logs missing or edited.
P4 Technical creativity (10) | Novel technique (e.g. multi-critic voting, auto A/B hooks, bilingual auto-dub, adaptive budget planner). | One thoughtful enhancement. | Standard tutorial-level pipeline. | Nothing beyond defaults.

9.3 Penalties

Violation | Penalty
Each fabricated fact / statistic | −10 (per instance)
Wrong helpline / contact in Category A or C | O4 = 0 and −5
Over API budget | −10
Late submission (per 10 min beyond 3 h) | −5
Video over 60 s or under 30 s | −5
Unlicensed music / copyrighted clips | −10 and asset must be replaced
Real person's face/voice generated | Output disqualified (Output = 0)
Pre-made script/video used in Live Run | Live Run = 0
Edited or fabricated logs | P1 and P3 = 0; academic-integrity referral

9.4 Awards (optional, announced at viva)

Principal's Choice — best admissions appeal (Category B/E)

Campus Guardian Award — most effective safety/wellbeing output (Category A)

Responsible AI Award — best Category C output

Best Autonomy — highest P1 with a working self-repair loop

Best Reproducible Workflow — cleanest repo, logs and one-command run

Best Fest Hype — best Category D output

9.5 Viva question bank

Walk us through your agent graph. Why does each agent exist?

Show a claim in the video and trace it to facts.json and the source page.

Where did the critic catch an error, and how did the pipeline fix it automatically?

Which stage consumed the most cost/time? How would you reduce it?

Re-run the Scriptwriter (or any stage) now with the same seed. What differs and why?

What would it take to reach the next autonomy level?

How do you prevent hallucinated helpline numbers or statistics?

What ethical risks did you identify for your topic, and how did you mitigate them?

10. Submission Checklist

- [ ] Final output named TeamID_BriefID_final.mp4 (or deployed URL + demo video) within specs
- [ ] .srt subtitle file and qa_report.json in outputs/
- [ ] Repo tagged v1.0-final; README with autonomy level and run command
- [ ] logs/run_log.jsonl, cost_report.csv, interventions.md complete for the Live Run
- [ ] prompts/ folder with all prompt templates
- [ ] assets/manifest.csv with source and licence for every asset
- [ ] Consent forms (if self-shot footage used)
- [ ] docs/agent_graph.png and docs/process_report.pdf
- [ ] No API keys or personal data committed

11. Appendices

11.1 Evaluator marking sheet (one per team)

Team ID: ____ Brief ID: ____ | Max | Awarded | Remarks
O1 Audience appeal / Clarity | 15
O2 Emotional impact | 10
O3 Narrative & quality | 10
O4 Accuracy | 5
P1 AI autonomy (level: ___ , interventions: ___) | 20
P2 Workflow design | 15
P3 Reproducibility (re-run: Y / N, cost: ₹___ , time: ___) | 15
P4 Technical creativity | 10
Penalties | — | –
Total (/100) | 100
Scaled lab marks | __ | Evaluator sign: _____

11.2 Human-intervention log format (interventions.md)

Time | Stage / agent | What the human changed | Reason
10:42 | Storyboarder | Replaced shot 4 prompt (generated image showed a real-looking face) | Integrity rule 7.1

11.3 Brief card template (print and cut)

BRIEF CARD — ID: ____ Category: _____ Topic:

_____ Audience: _____ Format: 9:16 / 16:9 / Digital Duration: ____ s Tone: _____ Language: _____ Must include: _____ Must avoid: _____ Call-to-action: _____

11.4 Practice brief (for Build phase dry-runs)

PRACTICE BRIEF — P00 Topic: "A day at the Agentic AI Lab" — what final-year students build Audience: second-year students choosing electives Format: 9:16, 40 s Tone: energetic, curious Must include: one real lab activity from the fact sheet Must avoid: invented project names or results CTA: "Explore the AI electives"